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CS 6001: Game Theory and Algorithmic Mechanism Design

Week 3

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Knowledge is the supreme goal



- ▶ Matrix games
- ▶ Relation between **maxmin** and PSNE
- ▶ Mixed Strategies
- ▶ Mixed Strategy Nash Equilibrium
- ▶ Find MSNE
- ▶ MSNE Characterization Theorem Proof
- ▶ Algorithm to find MSNE
- ▶ Existence of MSNE



Matrix games: *two player zero-sum* games

A special class with certain nice **security** and **stability** properties

Definition (Two player zero-sum games)

A NFG $\langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ with $N = \{1, 2\}$ and $u_1 + u_2 \equiv 0$

Question

Why called **matrix** game?

Answer

Possible to represent the game with only one matrix considering the utilities of player 1; player 2's utilities are negative of this matrix



Example: Penalty shoot game

Player 1

		Player 2	
		L	R
L	L	-1, 1	1, -1
	R	1, -1	-1, 1

$$\Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} =: u$$

Player 2's **maxmin** value is the **minmax** value of this matrix

		L	R	maxmin
Player 1	L	-1	1	-1
	R	1	-1	-1
	minmax	1	1	

Another example



		Player 2		
		L	C	R
Player 1	T	3, -3	-5, 5	-2, 2
	M	1, -1	4, -4	1, -1
	B	6, -6	-3, 3	-5, 5

		L	C	R	maxmin
Player 1	T	3	-5	-2	-5
	M	1	4	1	1
	B	6	-3	-5	-5
minmax		6	4	1	



Two examples together

Player 1

	L	R	maxmin
L	-1	1	-1
R	1	-1	-1
minmax	1	1	

Player 1

	L	C	R	maxmin
T	3	-5	-2	-5
M	1	4	1	1
B	6	-3	-5	-5
minmax	6	4	1	

Question

What are the PSNEs for the above games?

Answer

Left: no PSNE; **Right:** (M,R)

Saddle point



Saddle point of a matrix

The value is simultaneously the maximum in its column and minimum in its row i.e., maximum for player 1 and minimum for player 2

Question

What are the saddle points for the previous two games?

Saddle point



		L	R
Player 1	L	-1	1
	R	1	-1

		L	C	R
Player 1	T	3	-5	-2
	M	1	4	1
	B	6	-3	-5

Answer

For the first example: no saddle point, for the second: (M,R)

Theorem

In a matrix game with utility matrix u , (s_1^, s_2^*) is a saddle point iff it is a PSNE.*

Saddle point and PSNE



Proof.

Consider (s_1^*, s_2^*) to be a saddle point. By definition of saddle point, this happens iff $u(s_1^*, s_2^*) \geq u(s_1, s_2^*), \forall s_1 \in S_1$ and $u(s_1^*, s_2^*) \leq u(s_1^*, s_2), \forall s_2 \in S_2$. Since, $u \equiv u_1 \equiv -u_2$, the above is equivalent to $u_1(s_1^*, s_2^*) \geq u_1(s_1, s_2^*), \forall s_1 \in S_1$ and $u_2(s_1^*, s_2^*) \geq u_2(s_1^*, s_2), \forall s_2 \in S_2 \Leftrightarrow (s_1^*, s_2^*)$ is a PSNE. □

Consider **maxmin** and **minmax** values

$$\underline{v} = \max_{s_1 \in S_1} \min_{s_2 \in S_2} u(s_1, s_2)$$

maxmin

$$\bar{v} = \min_{s_2 \in S_2} \max_{s_1 \in S_1} u(s_1, s_2)$$

minmax

Question

How are the maxmin and minmax values related?

Relationship of the security values



Lemma

For matrix games $\bar{v} \geq \underline{v}$.

Proof.

$$u(s_1, s_2) \geq \min_{t_2 \in S_2} u(s_1, t_2), \quad \forall s_1, s_2,$$

definition of min

$$\Rightarrow \max_{t_1 \in S_1} u(t_1, s_2) \geq \max_{t_1 \in S_1} \min_{t_2 \in S_2} u(t_1, t_2), \quad \forall s_2 \in S_2$$

RHS was dominated for each s_1

$$\Rightarrow \min_{t_2 \in S_2} \max_{t_1 \in S_1} u(t_1, t_2) \geq \max_{t_1 \in S_1} \min_{t_2 \in S_2} u(t_1, t_2)$$

RHS was a constant





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Earlier examples and security values



		L	R	maxmin
Player 1	L	-1	1	-1
	R	1	-1	-1
	minmax	1	1	

$\bar{v} = 1 > -1 = \underline{v}$
PSNE does not exist

Earlier examples and security values (contd.)



		L	C	R	maxmin
Player 1	T	3	-5	-2	-5
	M	1	4	1	1
	B	6	-3	-5	-5
minmax		6	4	1	

$\bar{v} = 1 = \underline{v}$
PSNE exists



Define the following strategies

$$s_1^* \in \arg \max_{s_1 \in S_1} \min_{s_2 \in S_2} u(s_1, s_2),$$

maxmin strategy of player 1

$$s_2^* \in \arg \min_{s_2 \in S_2} \max_{s_1 \in S_1} u(s_1, s_2),$$

minmax strategy of player 2

Theorem

A game has a PSNE (equivalently, a saddle point) if and only if $\bar{v} = \underline{v} = u(s_1^, s_2^*)$, where s_1^* and s_2^* are maxmin and minmax strategies for players 1 and 2 respectively.*

Corollary: (s_1^*, s_2^*) is a PSNE

Proof of the PSNE Theorem



Proof

($\implies$) let (s_1^*, s_2^*) is a PSNE $\implies \bar{v} = \underline{v} = u(s_1^*, s_2^*)$ and s_1^* and s_2^* are maxmin and minmax
Since (s_1^*, s_2^*) is a PSNE, $u(s_1^*, s_2^*) \geq u(s_1, s_2^*), \forall s_1 \in S_1$.

$$\begin{aligned}\implies u(s_1^*, s_2^*) &\geq \max_{t_1 \in S_1} u(t_1, s_2^*) \\ &\geq \min_{t_2 \in S_2} \max_{t_1 \in S_1} u(t_1, t_2), \text{ since } s_2^* \text{ is a specific strategy} \\ &= \bar{v}\end{aligned}$$

Similarly, using the same argument for player 2, we get $\underline{v} \geq u(s_1^*, s_2^*)$

But $\bar{v} \geq \underline{v}$ (from the previous lemma), hence

$$\begin{aligned}u(s_1^*, s_2^*) &\geq \bar{v} \geq \underline{v} \geq u(s_1^*, s_2^*) \\ \implies u(s_1^*, s_2^*) &= \bar{v} = \underline{v}\end{aligned}$$

Also implies that the maxmin for 1 and minmax for 2 are s_1^* and s_2^* respectively.

Proof of the PSNE Theorem (contd.)



Proof (contd.)

($\Leftarrow$) i.e. $\bar{v} = \underline{v} = u(s_1^*, s_2^*)$ and s_1^* and s_2^* are maxmin and minmax $\implies (s_1^*, s_2^*)$ is a PSNE

$$\begin{aligned} u(s_1^*, s_2) &\geq \min_{t_2 \in S_2} u(s_1^*, t_2), \text{ by definition of min} \\ &= \max_{t_1 \in S_1} \min_{t_2 \in S_2} u(t_1, t_2), \text{ since } s_1^* \text{ is the maxmin strategy for player 1} \\ &= v \text{ (given)} \end{aligned}$$

Similarly, we can show $u(s_1, s_2^*) \leq v, \forall s_1 \in S_1$

But $v = u(s_1^*, s_2^*)$. Substitute and get that (s_1^*, s_2^*) is a PSNE.



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Mixed Strategies



Mixed strategy: *probability distribution* over the set of strategies of that player

		Player 2	
		$\frac{4}{5}$ L	$\frac{1}{5}$ R
Player 1	$\frac{2}{3}$ L	-1, 1	1, -1
	$\frac{1}{3}$ R	1, -1	-1, 1

- Consider a finite set A , define

$$\Delta A = \{p \in [0, 1]^{|A|} : \sum_{a \in A} p(a) = 1\}$$

- Mixed strategy set of player 1: $\Delta S_1 = \Delta\{L, R\}$, $(\frac{2}{3}, \frac{1}{3}) \in \Delta S_1$



Mixed Strategies (contd.)

- **Notation:** σ_i is a mixed strategy of player i
- $\sigma_i \in \Delta S_i$, i.e. , $\sigma_i : S_i \rightarrow [0, 1]$ s.t. $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$
- We are discussing non-cooperative games, the players choose their strategies independently
- The joint probability of player 1 picking s_1 and player 2 picking $s_2 = \sigma_1(s_1)\sigma_2(s_2)$
- Utility of player i at a mixed strategy profile (σ_i, σ_{-i}) is

$$u_i(\sigma_i, \sigma_{-i}) = \sum_{s_1 \in S_1} \sum_{s_2 \in S_2} \cdots \sum_{s_n \in S_n} \sigma_1(s_1) \cdot \sigma_2(s_2) \cdots \sigma_n(s_n) u_i(s_1, s_2, \dots, s_n)$$

- We are *overloading* u_i to denote the utility at *pure* and *mixed* strategies
- Utility at a mixed strategy is the **expectation** of the utilities at pure strategies; all the rules of expectation hold, e.g., linearity, conditional expectation, etc.

Example



		Player 2	
		$\frac{4}{5}$ L	$\frac{1}{5}$ R
Player 1	$\frac{2}{3}$ L	-1, 1	1, -1
	$\frac{1}{3}$ R	1, -1	-1, 1

$$u_1(\sigma_1, \sigma_2) = \frac{2}{3} \cdot \frac{4}{5} \cdot (-1) + \frac{2}{3} \cdot \frac{1}{5} \cdot (1) + \frac{1}{3} \cdot \frac{4}{5} \cdot (1) + \frac{1}{3} \cdot \frac{1}{5} \cdot (-1)$$



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Mixed Strategies Nash Equilibrium



Definition (Mixed Strategy Nash Equilibrium)

A **mixed strategy Nash equilibrium (MSNE)** is a mixed strategy profile $(\sigma_i^*, \sigma_{-i}^*)$, s.t.

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*), \quad \forall \sigma_i \in \Delta S_i \text{ and } \forall i \in N.$$

Question

Relation between **PSNE** and **MSNE**?

Answer

PSNE $\implies$ MSNE



An Alternative Definition

Theorem

A mixed strategy profile $(\sigma_i^*, \sigma_{-i}^*)$, is an **MSNE** if and only if $\forall s_i \in S_i$ and $\forall i \in N$

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(s_i, \sigma_{-i}^*).$$

Proof.

($\Rightarrow$): The pure strategy s_i is a special case of the mixed strategy, the mixed strategy with s_i having probability 1. Inequality holds by definition of MSNE

($\Leftarrow$) Pick an arbitrary mixed strategy σ_i of player i

$$\begin{aligned} u_i(\sigma_i, \sigma_{-i}^*) &= \sum_{s_i \in S_i} \sigma_i(s_i) \cdot u_i(s_i, \sigma_{-i}^*) \\ &\leq \sum_{s_i \in S_i} \sigma_i(s_i) \cdot u_i(\sigma_i^*, \sigma_{-i}^*) \\ &= u_i(\sigma_i^*, \sigma_{-i}^*) \cdot \sum_{s_i \in S_i} \sigma_i(s_i) = u_i(\sigma_i^*, \sigma_{-i}^*) \end{aligned}$$

Examples of MSNE



Question

Is the mixed strategy profile an **MSNE**?

		Player 2	
		$\frac{4}{5}$ L	$\frac{1}{5}$ R
Player 1	$\frac{2}{3}$ L	-1, 1	1, -1
	$\frac{1}{3}$ R	1, -1	-1, 1

- To answer this, we need to show that there does not exist any better mixed strategy for the player
- Expected utility of player 2 from L = $\frac{2}{3} \cdot 1 + \frac{1}{3} \cdot (-1) = \frac{1}{3}$
- Expected utility of player 2 from R = $\frac{2}{3} \cdot (-1) + \frac{1}{3} \cdot 1 = -\frac{1}{3}$



Examples of MSNE

Question

Is the mixed strategy profile an **MSNE**?

		Player 2	
		$\frac{1}{2}$ L	$\frac{1}{2}$ R
Player 1	$\frac{1}{2}$ L	-1, 1	1, -1
	$\frac{1}{2}$ R	1, -1	-1, 1

- Expected utility will increase if some probability is transferred from R to L
- $\Rightarrow$ the current profile is **not** an MSNE
- **Some balance in the utilities is needed**
- **Does there exist any improving mixed strategy?**



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How to find an MSNE

Definition (Support of mixed strategy/probability distribution)

For mixed strategy σ_i , the subset of strategy set of i on which σ_i has a positive mass is called the **support** of σ_i and is denoted by $\delta(\sigma_i)$. Formally, $\delta(\sigma_i) = \{s_i \in S_i : \sigma_i(s_i) > 0\}$.

Using the definition of support, here is a characterization of MSNE

Theorem

A mixed strategy profile $(\sigma_i^*, \sigma_{-i}^*)$ is an MSNE iff^a $\forall i \in N$

- 1 $u_i(s_i, \sigma_{-i}^*)$ is identical $\forall s_i \in \delta(\sigma_i^*)$,
- 2 $u_i(s_i, \sigma_{-i}^*) \geq u_i(s'_i, \sigma_{-i}^*), \forall s_i \in \delta(\sigma_i^*), s'_i \notin \delta(\sigma_i^*)$.

^aThis is a shorthand for 'if and only if'.

Implication



Consider Penalty Shoot Game

		Goalkeeper	
		L	R
Shooter	L	-1, 1	1, -1
	R	1, -1	-1, 1

Case 1: support profile $(\{L\}, \{L\})$: for player 1, $s'_1 = R$ – violates condition 2

Case 2: support profile $(\{L, R\}, \{L\})$ – symmetric for the other case

For Player 1, the expected utility has to be the same for L and R - **not possible** – violates condition 1

Implication



Case 3: support profile $(\{L, R\}, \{L, R\})$: condition 2 is vacuously satisfied

For condition 1, let player 1 chooses L w.p. p and player 2 choose L w.p. q

For player 1:

$$u_1(L, (q, 1 - q)) = u_1(R, (q, 1 - q)) \Rightarrow (-1)q + 1 \cdot (1 - q) = 1 \cdot q + (-1)(1 - q) \Rightarrow q = \frac{1}{2}$$

For player 2:

$$u_2((p, 1 - p), L) = u_2((p, 1 - p), R) \Rightarrow p = \frac{1}{2}$$

MSNE =

$$\left(\left(\frac{1}{2}, \frac{1}{2} \right), \left(\frac{1}{2}, \frac{1}{2} \right) \right)$$

Exercises



		Player 2	
		F	C
Player 1	F	2,1	0,0
	C	0,0	1,2

		Player 2		
		F	C	D
Player 1	F	2,1	0,0	1,1
	C	0,0	1,2	2,0



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MSNE Characterization Theorem



Theorem

A mixed strategy profile is an MSNE iff $\forall i \in N$

- 1 $u_i(s_i, \sigma_{-i}^*)$ is identical $\forall s_i \in \delta(\sigma_i^*)$,
- 2 $u_i(s_i, \sigma_{-i}^*) \geq u_i(s'_i, \sigma_{-i}^*), \forall s_i \in \delta(\sigma_i^*), s'_i \notin \delta(\sigma_i^*)$.

Observations:

- $\max_{\sigma_i \in \Delta S_i} u_i(\sigma_i, \sigma_{-i}) = \max_{s_i \in S_i} u_i(s_i, \sigma_{-i})$
maximizing w.r.t. a distribution $\Leftrightarrow$ whole probability mass at max
- If $(\sigma_i^*, \sigma_{-i}^*)$ is an MSNE, then

$$\max_{\sigma_i \in \Delta S_i} u_i(\sigma_i, \sigma_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^*) = \max_{s_i \in \delta(\sigma_i^*)} u_i(s_i, \sigma_{-i}^*)$$

the maximizer must lie in $\delta(\sigma_i^*)$ – if not, then put all probability mass on that $s'_i \notin \delta(\sigma_i^*)$ that has the maximum value of the utility – $(\sigma_i^*, \sigma_{-i}^*)$ is not a MSNE

Proof of MSNE Characterization Theorem



($\Rightarrow$) Given $(\sigma_i^*, \sigma_{-i}^*)$ is an MSNE

$$u_i(\sigma_i^*, \sigma_{-i}^*) = \max_{\sigma_i \in \Delta S_i} u_i(\sigma_i, \sigma_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^*) = \max_{s_i \in \delta(\sigma_i^*)} u_i(s_i, \sigma_{-i}^*) \quad (1)$$

By definition of expected utility

$$u_i(\sigma_i^*, \sigma_{-i}^*) = \sum_{s_i \in S_i} \sigma_i^*(s_i) u_i(s_i, \sigma_{-i}^*) = \sum_{s_i \in \delta(\sigma_i^*)} \sigma_i^*(s_i) u_i(s_i, \sigma_{-i}^*) \quad (2)$$

Equations (1) and (2) are equal, i.e., max is equal to positive weighted average – **can happen only when all values are same: proves condition 1**

Proof (contd.)



For **condition 2**: Suppose for contradiction, there exists $s_i \in \delta(\sigma_i^*)$ and $s'_i \notin \delta(\sigma_i^*)$ s.t.
 $u_i(s_i, \sigma_{-i}^*) < u_i(s'_i, \sigma_{-i}^*)$

We can shift the probability mass $\sigma^*(s_i)$ to s'_i , this new mixed strategy gives a strict higher utility to player i : contradicts MSNE

This completes the proof of the necessary direction.

($\Leftarrow$) Given the 2 conditions of the theorem, need to show that $(\sigma_i^*, \sigma_{-i}^*)$ is an MSNE

$$\text{Let } u_i(s_i, \sigma_{-i}^*) = m_i(\sigma_{-i}^*), \forall s_i \in \delta(\sigma_i^*)$$

condition 1

$$\text{Note } m_i(\sigma_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^*)$$

condition 2



$$\begin{aligned} u_i(\sigma_i^*, \sigma_{-i}^*) &= \sum_{s_i \in \delta(\sigma_i^*)} \sigma_i^*(s_i) u_i(s_i, \sigma_{-i}^*), \\ &= m_i(\sigma_{-i}^*) \\ &= \max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^*) \\ &= \max_{\sigma_i \in \Delta S_i} u_i(\sigma_i, \sigma_{-i}^*) \\ &\geq u_i(\sigma_i, \sigma_{-i}^*), \quad \forall \sigma_i \in \Delta S_i \end{aligned}$$

by definition of $\delta(\sigma_i^*)$

previous conclusion

previous conclusion

from the observation

This proves the sufficient direction. The result yields an algorithmic way to find MSNE



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MSNE characterization theorem to algorithm

Consider a NFG $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$

The total number of supports of $S_1 \times S_2 \times S_3 \cdots \times S_n$ is

$$K = (2^{|S_1|} - 1) \times (2^{|S_2|} - 1) \times \cdots \times (2^{|S_n|} - 1)$$

For every support profile $X_1 \times X_2 \times \cdots \times X_n$, where $X_i \subseteq S_i$, solve the following feasibility program

Program

$$\begin{aligned} w_i &= \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \neq i} \sigma_j(s_j) \right) \cdot u_i(s_i, s_{-i}), \forall s_i \in X_i, \forall i \in N \\ w_i &\geq \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \neq i} \sigma_j(s_j) \right) \cdot u_i(s_i, s_{-i}), \forall s_i \in S_i \setminus X_i, \forall i \in N \\ \sigma_j(s_j) &\geq 0, \forall s_j \in S_j, \forall j \in N, \quad \sum_{s_j \in X_j} \sigma_j(s_j) = 1, \forall j \in N \end{aligned}$$

Remarks on the algorithm



Program

$$\begin{aligned}w_i &= \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \neq i} \sigma_j(s_j) \right) \cdot u_i(s_i, s_{-i}), \forall s_i \in X_i, \forall i \in N \\w_i &\geq \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \neq i} \sigma_j(s_j) \right) \cdot u_i(s_i, s_{-i}), \forall s_i \in S_i \setminus X_i, \forall i \in N \\ \sigma_j(s_j) &\geq 0, \forall s_j \in S_j, \forall j \in N, \quad \sum_{s_j \in X_j} \sigma_j(s_j) = 1, \forall j \in N\end{aligned}$$

- This is not a linear program unless $n = 2$
- For general game, there is no poly-time algorithm
- Problem of finding an MSNE is PPAD-complete [Polynomial Parity Argument on Directed graphs]¹

¹Daskalakis, Goldberg, Papadimitriou, "The Complexity of Computing a Nash Equilibrium" [2009]

MSNE and Dominance



The previous algorithm can be applied to a smaller set of strategies by removing the dominated strategies

Is there a dominated strategy in this game? Domination can be via mixed strategies too

		Player 2	
		L	R
Player 1	T	4, 1	2, 5
	M	1, 3	6, 2
	B	2, 2	3, 3



Theorem

If a pure strategy s_i is strictly dominated by a mixed strategy $\sigma_i \in \Delta S_i$, then in every MSNE of the game, s_i is chosen with probability zero.

So, We can remove such strategies without loss of equilibrium



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Definition (Finite Games)

A game is said to be **finite** when the number of players is finite, and each player has a finite set of strategies.

Theorem (Nash 1951)

Every finite game has a (mixed) Nash equilibrium.

Proof requires a few tools and a result from real analysis. Proof is separately given in the course webpage.



Existence of MSNE

Some background for understanding the proof.

- A set $S \subseteq \mathbb{R}^n$ is **convex** if $\forall x, y \in S$ and $\forall \lambda \in [0, 1]$, $\lambda x + (1 - \lambda)y \in S$.
- A set $S \subseteq \mathbb{R}^n$ is **closed** if it contains all its **limit points** (points whose **every** neighborhood contains a point in S). Example of a set that is **not closed**: $[0, 1)$ - every ball of radius $\epsilon > 0$ around 1 has a member of $[0, 1)$, but 1 is not in the set $[0, 1)$.
- A set $S \subseteq \mathbb{R}^n$ is **bounded** if $\exists x_0 \in \mathbb{R}^n$ and $R \in (0, \infty)$ s.t. $\forall x \in S$, $\|x - x_0\|_2 < R$.
- A set $S \subseteq \mathbb{R}^n$ is **compact** if it is **closed** and **bounded**.

A result from real analysis (proof omitted):

Brouwer's fixed point theorem

If $S \subseteq \mathbb{R}^n$ is **convex** and **compact** and $T : S \rightarrow S$, is **continuous** then T has a fixed point, i.e., $\exists x^* \in S$ s.t. $T(x^*) = x^*$.



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